

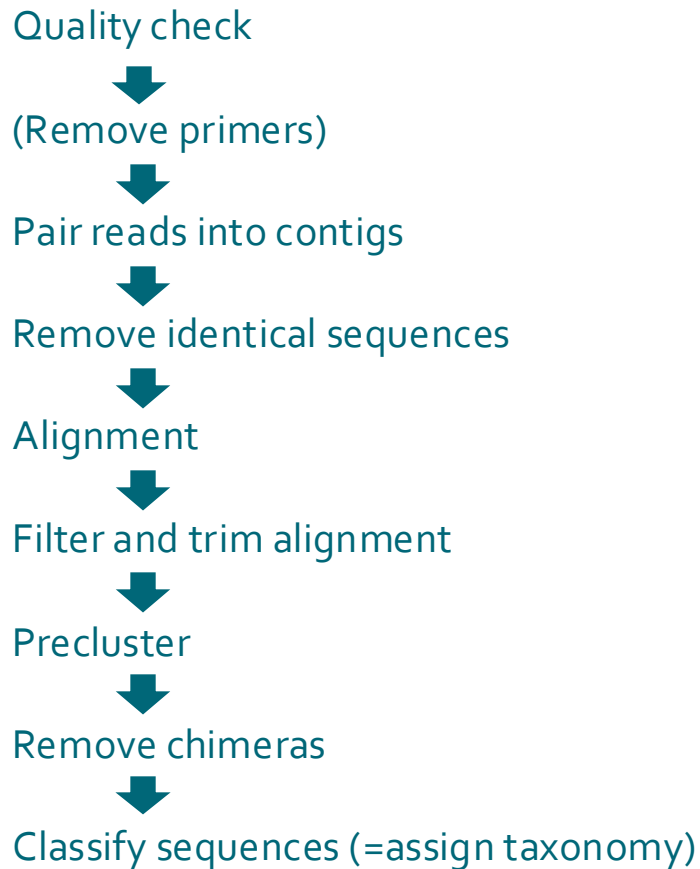


Microbial community / environmental DNA analysis with Chipster (Day 2)

Heli Juottonen and Eija Korpelainen (slides modified from: Jesse Harrison)
CSC – IT Center for Science Ltd.
chipster@csc.fi



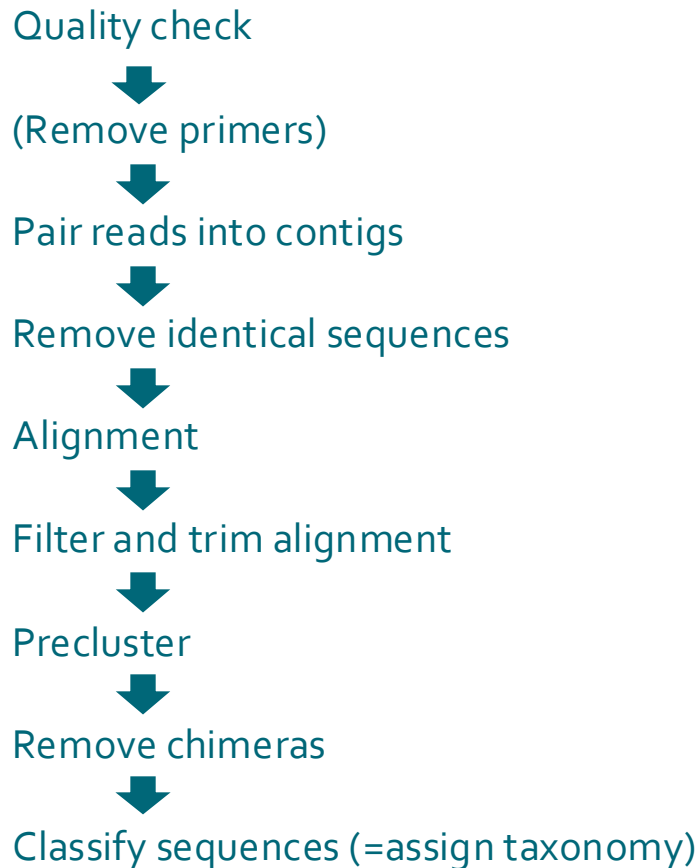
Outline of Day 1: Preprocessing reads



Output so far:

1. FASTA file of processed reads
2. count file (which read in which sample)
3. taxonomy file (taxonomy of each read)

Outline of Day 1: Preprocessing reads



Output so far:

1. FASTA file of processed reads
2. count file (which read in which sample)
3. taxonomy file (taxonomy of each read)



Day 2: Community analysis

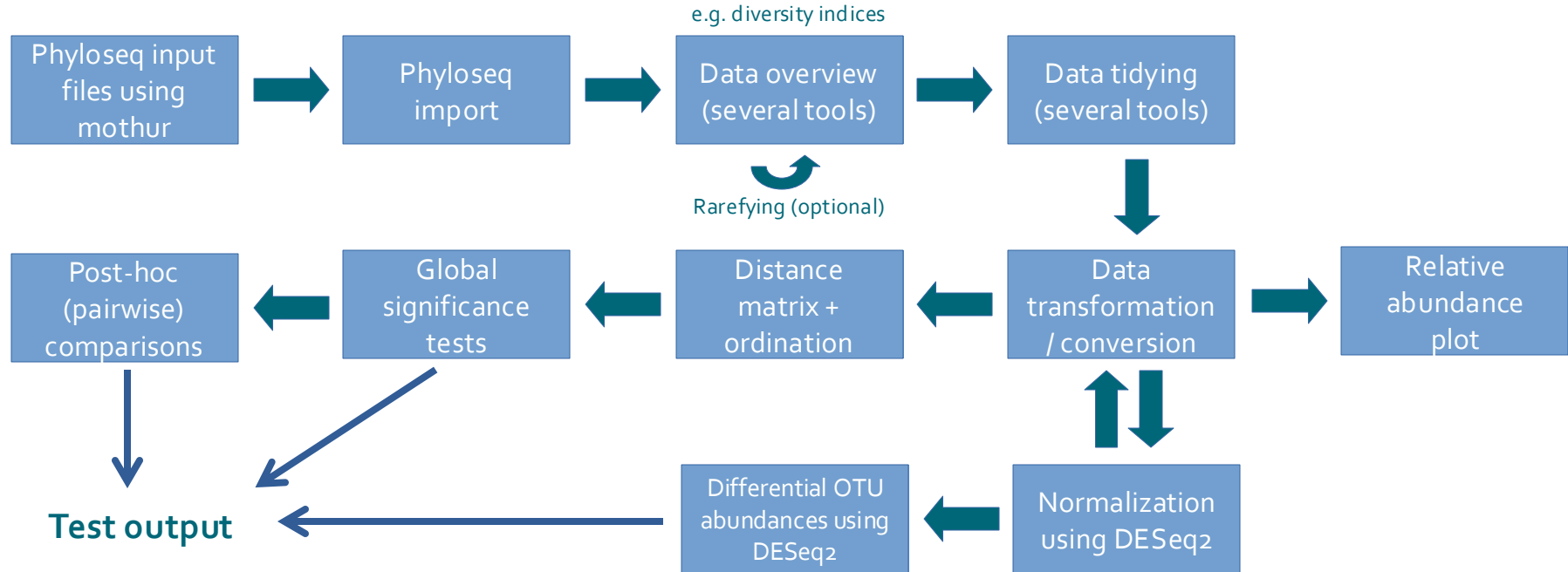
- Clustering into OTUs
- Import into **Phyloseq**
- Data tidying & transformations
- Taxonomy plots
- Alpha diversity
- Beta diversity: ordinations & statistics



Part 1: Tool overview and data importing



Workflow for microbial community analysis in Chipster



Cluster sequences to OTUs and classify them

Cluster sequences to OTUs and classify them

✕

Parameters

Reset All

Type of data

16S or 18S

Indicate if you have ITS data as it is treated differently.

Clustering cutoff

0.03

Dissimilarity threshold for OTU clustering, e.g. a cut-off value of 0.03 corresponds to 97% similarity

Input files

FASTA file

chimeras.removed.fasta.gz

Mothur count file

chimeras.removed.count_table

Sequences taxonomy assignment file

sequences-taxonomy-assignmer

Cluster sequences to OTUs and classify them

Specifications :

- type of data (16S/18S or ITS)
- % cut off for OTU clustering
- files produced by mothur
 - final FASTA
 - count file
 - taxonomy file = taxonomy assignment of each read

Cluster sequences to OTUs and classify them

Parameters

Type of data
16S or 18S

Indicate if you have ITS data as it is treated differently.

Clustering cutoff
0.03

Dissimilarity threshold for OTU clustering, e.g. a cut-off value of 0.03 corresponds to 97% similarity

Input files

FASTA file
chimeras.removed.fasta.gz

Mothur count file
chimeras.removed.count_table

Sequences taxonomy assignment file
sequences-taxonomy-assignmer

Reset All

Cluster sequences to OTUs and classify them

Generated output files:

- .shared file (mothur file format)
 - samples in rows, OTUs in columns
 - how many reads of each OTU in each sample (OTU table)
- consensus taxonomy file
 - taxonomy assignments of OTUs
- phenodata file

file.opti_mcc.shared ...

Spreadsheet Text Phenodata Open in New Tab Details

Showing all 19 rows.

label	Group	numOtu	Otu001	Otu002	Otu003	Otu004	Otu005	Otu006	Otu007	Otu008	Otu009
0.03	F3D0	438	471	307	433	659	369	372	169	173	145
0.03	F3D1	438	338	313	67	72	171	121	130	181	95
0.03	F3D141	438	373	338	492	446	255	296	214	343	124
0.03	F3D142	438	233	271	156	260	123	196	207	82	69
0.03	F3D143	438	174	153	221	321	159	194	125	96	64
0.03	F3D144	438	317	243	340	491	231	291	138	38	142
0.03	F3D145	438	471	430	552	686	374	499	373	133	222
0.03	F3D146	438	258	202	386	416	204	267	203	69	86
0.03	F3D147	438	1235	1100	1100	1698	724	1116	564	91	254
0.03	F3D148	438	699	668	890	1278	485	764	521	563	193
0.03	F3D149	438	730	719	908	924	589	704	478	574	267

file.opti_mcc.0.03.cons.taxonomy ...

Spreadsheet Text Open in New Tab Details

Showing the first 100 of 438 rows. View in [full screen](#) to see all rows.

OTU	Size	Taxonomy
Otu001	11512	Bacteria(100);Bacteroidota(100);Bacteroidia(100);Bacteroidales(100);Muribaculaceae(100)
Otu002	8946	Bacteria(100);Bacteroidota(100);Bacteroidia(100);Bacteroidales(100);Muribaculaceae(100)
Otu003	7846	Bacteria(100);Bacteroidota(100);Bacteroidia(100);Bacteroidales(100);Muribaculaceae(100)
Otu004	7759	Bacteria(100);Bacteroidota(100);Bacteroidia(100);Bacteroidales(100);Muribaculaceae(100)
Otu005	7067	Bacteria(100);Bacteroidota(100);Bacteroidia(100);Bacteroidales(100);Muribaculaceae(100)
Otu006	6985	Bacteria(100);Bacteroidota(100);Bacteroidia(100);Bacteroidales(100);Muribaculaceae(100)
Otu007	6550	Bacteria(100);Bacteroidota(100);Bacteroidia(100);Bacteroidales(100);Bacteroidaceae(100)
Otu008	5615	Bacteria(100);Bacteroidota(100);Bacteroidia(100);Bacteroidales(100);Rikenellaceae(100);A
Otu009	3775	Bacteria(100);Bacteroidota(100);Bacteroidia(100);Bacteroidales(100);Muribaculaceae(100)

Phenodata file: fill in sample information

file.opti_mcc.shared ...

Spreadsheet Text **Phenodata** Open in New Tab

+ Add column

sample	original_name	group	×	time	×
F3D0		a		early	
F3D1		a		early	
F3D141		a		late	
F3D142		a		late	
F3D143		a		late	
F3D144		a		late	
F3D145		a		late	
F3D146		a		late	
F3D147		a		late	
F3D148		b		late	
F3D149		b		late	
F3D150		b		late	
F3D2		b		early	
F3D3		b		early	
F3D5		b		early	
F3D6		b		early	
F3D7		b		early	
F3D8		b		early	
F3D9		b		early	

The phenodata file is an editable table with

- 1) unique IDs for each sample
- 2) information on sample groupings

Converting input files into a phyloseq object

Phyloseq is a multi-use R package for microbial community data processing and analysis

<https://joey711.github.io/phyloseq/>

X

Convert Mothur files into phyloseq object

Parameters

Reset All

Phenodata variable with sequencing sample IDs

sample

Phenodata variable with unique IDs for each community profile.

Input files

Mothur shared file

file.opti_mcc.shared

Mothur constaxonomy file

file.opti_mcc.0.03.cons.taxonomy

Phenodata

Using phenodata of *file.opti_mcc.shared*

Converting input files into a phyloseq object

```
### Imported phyloseq object ###

phyloseq-class experiment-level object
otu_table() OTU Table:      [ 438 taxa and 19 samples ]
sample_data() Sample Data:  [ 19 samples by 4 sample variables ]
tax_table()  Taxonomy Table: [ 438 taxa by 6 taxonomic ranks ]

### Sample names ###

[1] "F3D0" "F3D1" "F3D141" "F3D142" "F3D143" "F3D144" "F3D145" "F3D146"
[9] "F3D147" "F3D148" "F3D149" "F3D150" "F3D2" "F3D3" "F3D5" "F3D6"
[17] "F3D7" "F3D8" "F3D9"

### Sample variables ###

[1] "sample"      "original_name" "group"      "time"
```

Produces a **phyloseq object (.Rda)** and a text summary

The Rda file is used as **the input for downstream analyses**

OTU table, taxonomy table and sample data can be exported for use outside Chipster (**Microbial amplicon data analyses / Extract information from the Phyloseq object**)



Part 2: Data tidying and alpha diversity



Taxon-level clean-up tools

Under **Microbial amplicon data analyses**:

- **Filter by taxonomic group**
 - Remove non-specific sequences (keep e.g. Bacteria or Archaea only)
- **Remove selected taxa**
 - Remove chloroplast and/or mitochondrial sequences
 - (Manually remove specific taxa)
- **Overview of taxon composition**
 - user-specified level

Remove selected taxa

Parameters

Remove class Chloroplast

Remove class Chloroplast

yes

Remove family Mitochondria

Remove family Mitochondria

yes

Level of biological organization for manual taxon removal

Select the desired taxonomic level; default is phylum

Phylum

Taxon to be removed

Name of taxon to be filtered out

2nd taxon to be removed

Name of taxon to be filtered out

3rd taxon to be removed

Name of taxon to be filtered out

4th taxon to be removed

Name of taxon to be filtered out

5th taxon to be removed

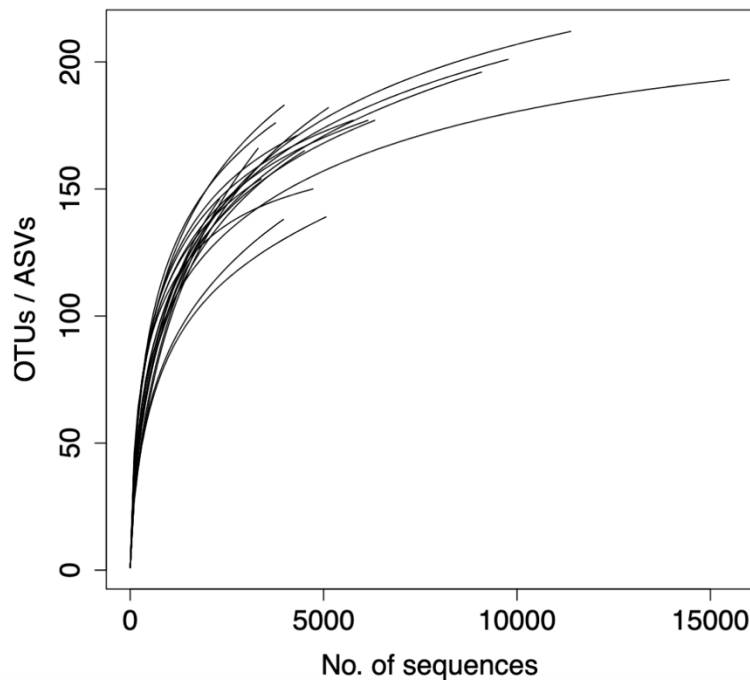
Name of taxon to be filtered out

Input files

Phyloseq object in .Rda format

ps.Rda

Data inspection: Sequence numbers, rarefaction curve and alpha diversity estimates



Per-sample sequence no.s

F3D0	F3D1	F3D141	F3D142	F3D143	F3D144	F3D145	F3D146	F3D147	F3D148	F3D149
6149	4727	4504	2311	2347	3307	5125	3756	11390	9085	9770
F3D150	F3D2	F3D3	F3D5	F3D6	F3D7	F3D8	F3D9			
3981	15481	5062	3387	6323	3962	4306	5758			

Alpha diversity estimates (observed OTUs, Chao1, Shannon's index, Pielou's evenness)

	Observed	Chao1	se.chao1	Shannon	pielou	time
F3D0	177	210.0000	15.261725	3.853259	0.7444256	early
F3D1	150	163.1538	7.993669	4.032404	0.8047691	early
F3D141	165	194.6897	12.042909	3.499614	0.6853998	late
F3D142	146	193.0400	17.787187	3.343976	0.6709952	late
F3D143	143	177.7308	13.935854	3.481141	0.7014407	late
F3D144	166	246.6400	27.261107	3.212135	0.6283534	late
F3D145	182	231.2143	17.858418	3.160095	0.6072427	late
F3D146	176	219.1579	17.951815	3.758571	0.7269283	late

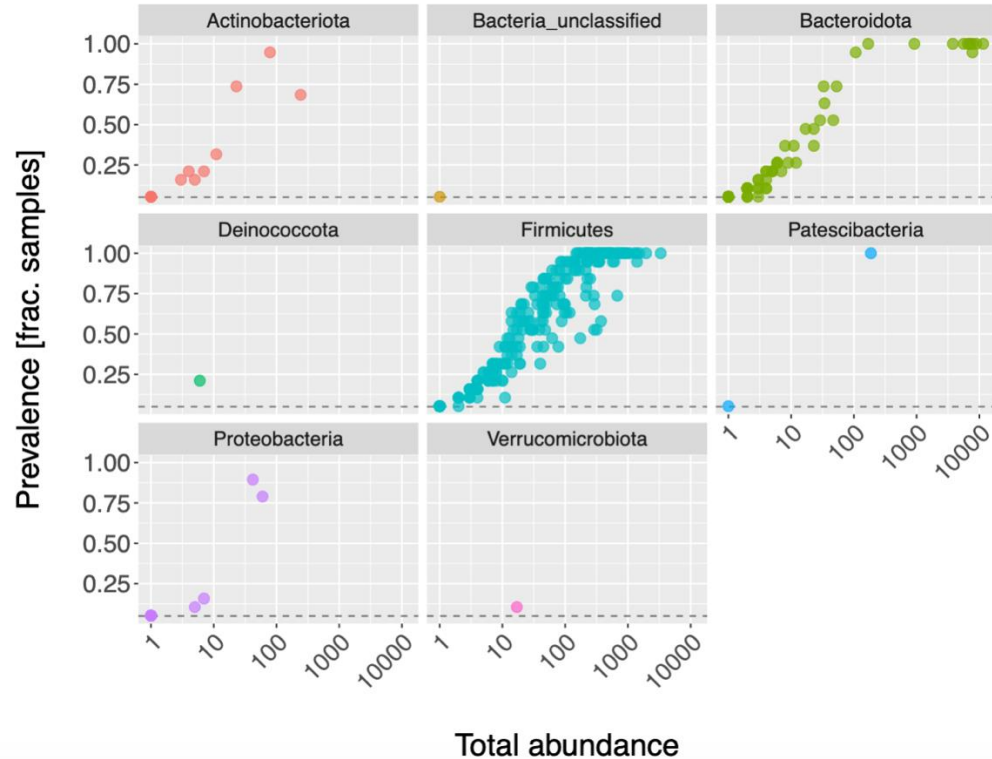
Filtering low-abundance OTUs

- Filter out OTUs that occur in less than x % of samples
 - Proportional prevalence filtering
- Remove singletons and doubletons
 - Remove OTUs with 0-2 occurrences

Visualizing low-abundance OTUs

Additional prevalence summaries

- Visualization of OTU prevalence at phylum level
- Text summary on low prevalence OTUs



Alpha diversity

Diversity within a habitat: how many species are present, at which relative amounts?

- Species richness - how many species?
 - observed number of OTUs
 - richness estimators (Chao1)
- Evenness – abundance distribution of species?
 - all OTUs equally abundant vs. a few dominant and a lot of rare OTUs
 - Pielou's evenness
- Many diversity indices **combine richness and evenness** (e.g. Shannon index)

Alpha diversity

Diversity within a habitat: how many species are present, at which relative amounts?

- Species richness - how many species?

- observed number of OTUs
- richness estimators (Chao1)

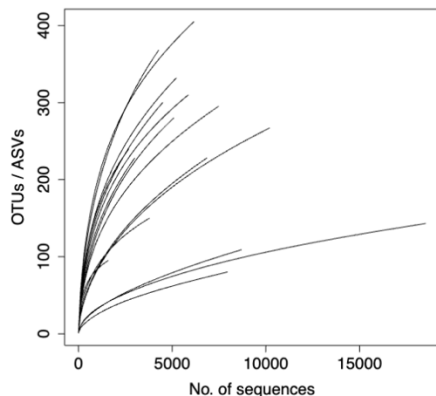
note: do not use after removing singletons and doubletons!

- Evenness – abundance distribution of species?

- all OTUs equally abundant vs. a few dominant and a lot of rare OTUs
- Pielou's evenness

- Many diversity indices **combine richness and evenness** (e.g. Shannon index)

Rarefaction?



Uneven sequence numbers among samples can bias comparisons, especially with **alpha diversity**. Solutions:

- rarefying: equal number of reads picked from all samples (Rarefy OTU data to even depth)
- data transformation (Transform OTU counts)

PLOS COMPUTATIONAL BIOLOGY

OPEN ACCESS PEER-REVIEWED
RESEARCH ARTICLE

Waste Not, Want Not: Why Rarefying Microbiome Data Is Inadmissible

Paul J. McMurdie, Susan Holmes

Published: April 3, 2014 • <https://doi.org/10.1371/journal.pcbi.1003531>

Waste not, want not: Revisiting the analysis that called into question the practice of rarefaction

Patrick D. Schloss

doi: <https://doi.org/10.1101/2023.06.23.546312>

Rarefaction is currently the best approach to control for uneven sequencing effort in amplicon sequence analyses

Patrick D. Schloss

doi: <https://doi.org/10.1101/2023.06.23.546313>



Part 3: Transformations and ordinations



Transformation of OTU data

Four options (April 2025)

Transform OTU counts

Parameters

Data treatment

Choice between data transformation types

✓ Centered log-ratio transformation with pseudocount

Relative abundances (%)

Hellinger transformation

DESeq2 format conversion and variance-stabilizing transformation

Reset All

Phenodata variable used for DESeq2 conversion

Select a phenodata variable used to specify the experimental design when converting the data to DESeq2 format.

Input files

Phyloseq object in Rda format

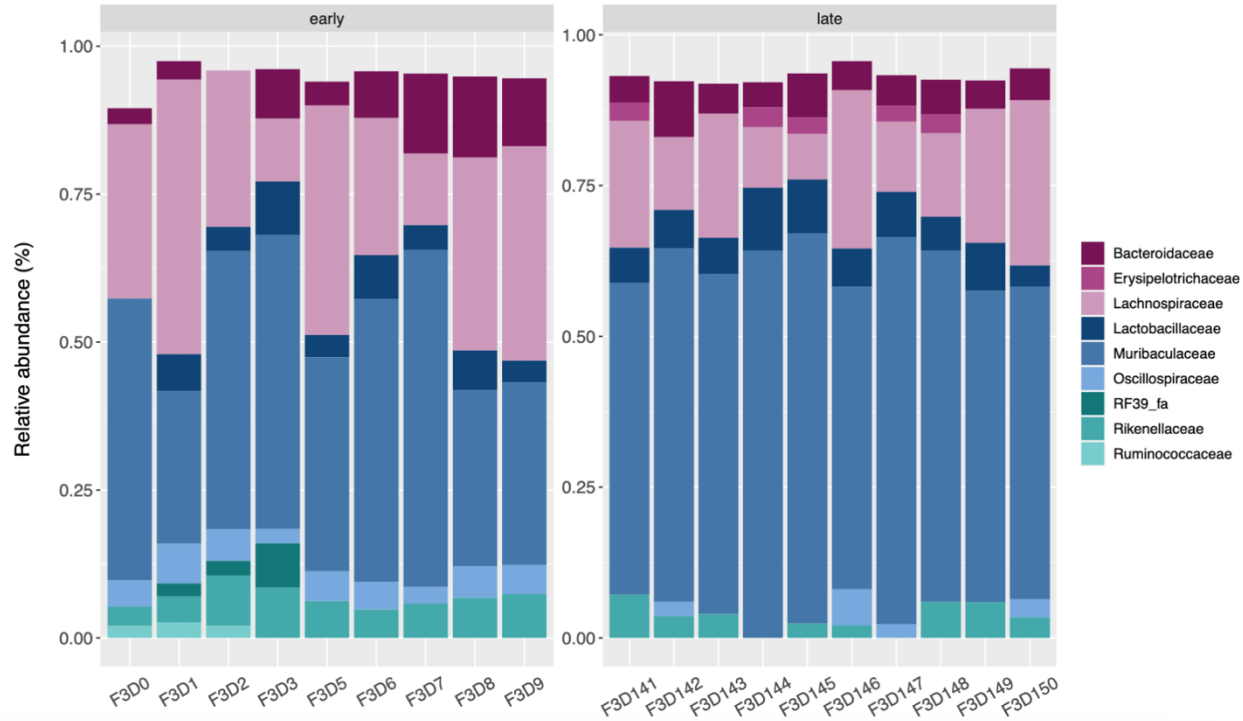
ps_ind.Rda

Phenodata

Using phenodata of *file.opti_mcc.shared*

21

Relative abundance (%) bar plots



Beta diversity

Change in community composition among habitats

- Does the microbial community composition in treatment A differ from treatment B?
- Unlike in alpha diversity, OTU identity matters
- Both identity and relative abundance of OTUs usually included
- Quantified with **distance or dissimilarity measures**

Distance matrices and ordinations

Distance measures available: **Bray-Curtis** or Euclidean

- centered log ratio (CLR) transformation + Euclidean distance = **Aitchinson distance**

Ordinations: visualizing beta diversity

- nMDS (non-metric multidimensional scaling)
 - overall variation among samples displayed
- db-RDA (distance-based redundancy analysis)
 - focus on the variation explained by phenodata variable(s)

Distance matrices and ordinations

Parameters
Reset All

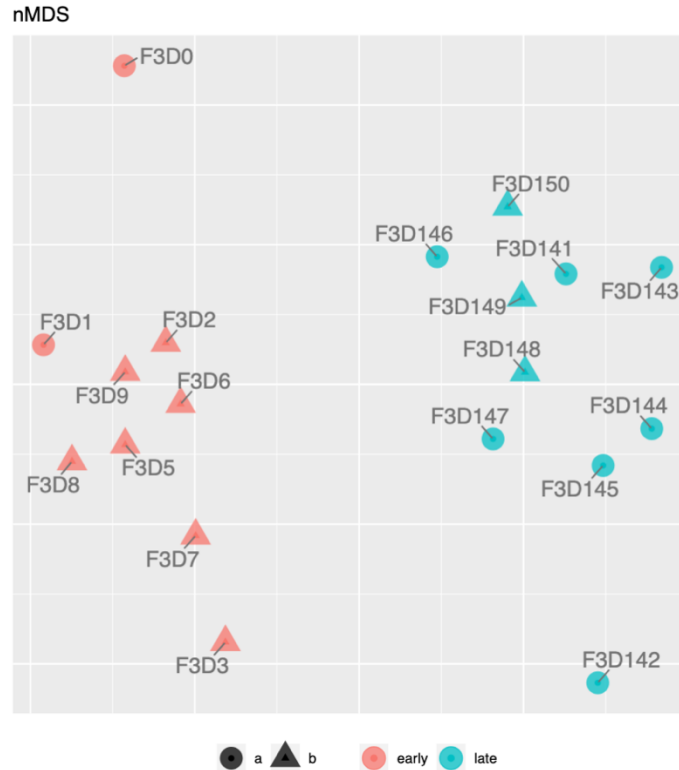
Type of distance measure
Bray-Curtis
Choice between Euclidean and Bray-Curtis distances

Type of ordination
nMDS
Choice between using non-metric multidimensional scaling (nMDS) or distance-based redundancy analysis (db-RDA)

Phenodata variable with sequencing sample IDs
sample
Phenodata variable with unique IDs for each community profile.

Recommended for more information:
 Guide to Statistical Analysis in Microbial Ecology:
<https://sites.google.com/site/mb3gustame/>

Non-metric multidimensional scaling (nMDS)



Distance-based redundancy analysis (db-RDA)

Requires specifying one or more phenodata variables

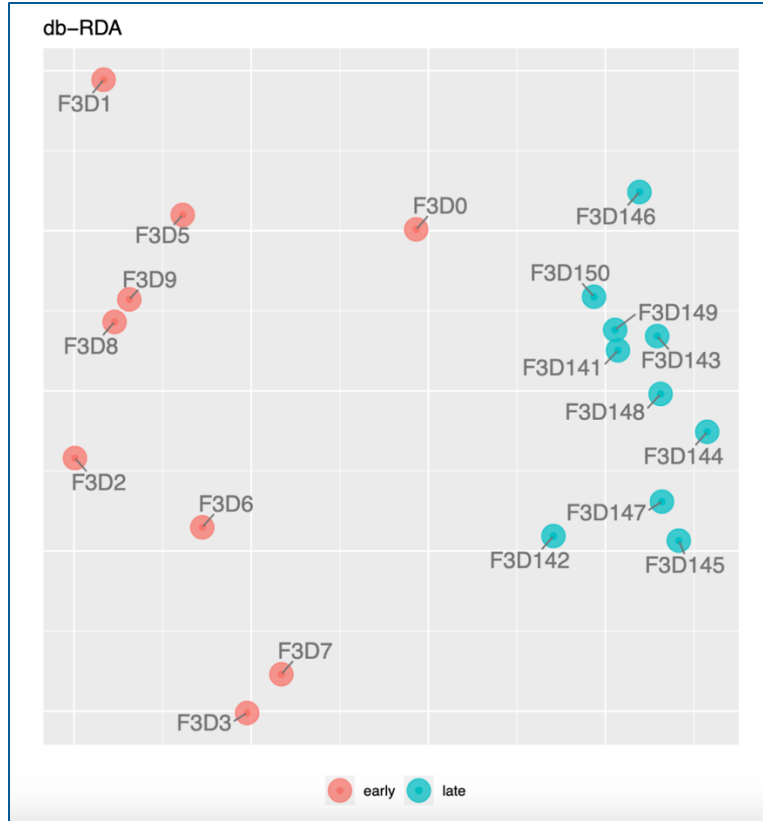


Distance matrices and ordinations

Parameters Reset All

Type of distance measure Choice between Euclidean and Bray-Curtis distances	Bray-Curtis	
Type of ordination Choice between using non-metric multidimensional scaling (nMDS) or distance-based redundancy analysis (db-RDA)	db-RDA	
Phenodata variable with sequencing sample IDs Phenodata variable with unique IDs for each community profile.	sample	
Show sample IDs in ordination? Should sample labels be plotted next to data points in the ordination?	Yes	
Phenodata variable for grouping ordination points by colour Phenodata variable used for grouping ordination points by colour.	time	
Phenodata variable for grouping ordination points by shape Phenodata variable used for grouping ordination points by shape.		
Phenodata variable 1 for db-RDA formula specification 1st phenodata variable used in the "formula" argument when performing db-RDA (minimum requirement is 1 variable)	time	
Phenodata variable 2 for db-RDA formula specification 2nd phenodata variable used in the "formula" argument when performing db-RDA		
Phenodata variable 3 for db-RDA formula specification 3rd phenodata variable used in the "formula" argument when performing db-RDA		

Distance-based redundancy analysis (db-RDA)



Constrained ordination =
focus on the community
variation explained by the
phenodata variable(s)

Recommended for more information:
Guide to Statistical Analysis in Microbial Ecology:
<https://sites.google.com/site/mb3gustame/>



Part 4: Statistics



PERMANOVA (permutational multivariate analysis of variance)

Input: distance matrix (ps_dist.Rda)

- Global test: **'Does community structure differ between sample groups?'**
 - Pairwise test: 'Which groups differ from one another?'
- Currently: several phenodata variables -> added sequentially -> order matters!
- Influenced by both **location** and **dispersion**

Global PERMANOVA for OTU abundance data

Parameters
Reset All

Main effects only vs. main effects and interactions?
Main effects only

PERMANOVA formula to be used

Phenodata variable 1
time

First phenodata variable used to specify the formula to be tested

Phenodata variable 2

Second phenodata variable used to specify the formula to be tested

Phenodata variable 3

Third phenodata variable used to specify the formula to be tested

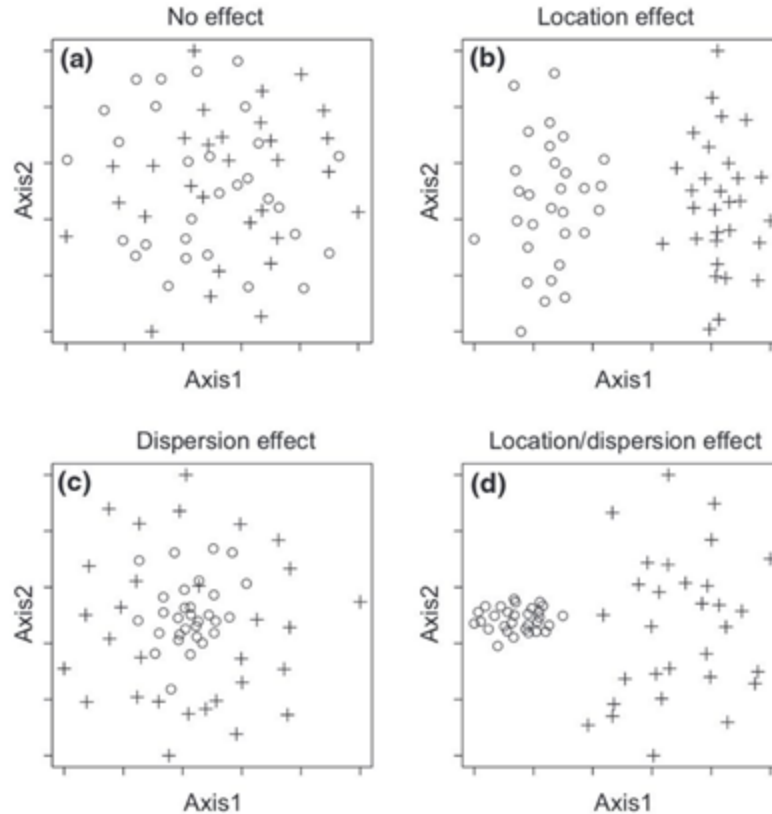
Input files

Data set in Rda format
ps_dist.Rda

Location vs. dispersion

A significant PERMANOVA result can be due to:

- Location effect
- Dispersion effect
- Combination of both



PERMANOVA output

```
### Global PERMANOVA summary ###
```

```
$aov.tab
```

```
Permutation: free
```

```
Number of permutations: 999
```

```
Terms added sequentially (first to last)
```

	Df	SumsOfSqs	MeanSqs	F.Model	R2	Pr(>F)
get(pheno1)	1	2399.5	2399.54	8.6879	0.33821	0.001 ***
Residuals	17	4695.3	276.19		0.66179	
Total	18	7094.8			1.00000	

```
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
$call
```

```
adonis(formula = ps_dist ~ get(pheno1), data = ps_df)
```

Df: Degrees of freedom

F.Model: Test statistic (pseudo- F)

R2: Variation explained by the model

Pr(>F): Statistical significance (p value)

PERMDISP: test for the homogeneity of multivariate dispersion

Input: distance matrix

- Test if a significant PERMANOVA result is due to dispersion, not (only) location
- Significant result -> be careful with PERMANOVA interpretation

PERMDISP for OTU abundance data

Parameters

Phenodata variable 1
Phenodata variable used for first PERMDISP analysis

time

Reset All

Phenodata variable 2
Phenodata variable used for second PERMDISP analysis

Phenodata variable 3
Phenodata variable used for third PERMDISP analysis

Input files

Data set in .Rda format

ps_dist.Rda

Post-hoc comparisons

Sample group comparisons **following significant global PERMANOVA:**

- Pairwise PERMANOVA (similar as global test but for sample pairs)

Dispersion comparisons **following significant PERMDISP:**

- Tukey's Honestly Significant Difference (HSD) test
- Both methods use a correction for multiple testing (Benjamin-Hochberg correction)

DESeq2

- Originates from the RNA-seq field
- Addresses the question: **'Which taxa are differentially abundant between sample groups?'**
- Enables inferences such as: 'Illness x is associated with a reduction in the abundance of beneficial gut microbes y and z'
- Input: untransformed data -> convert to DESeq2 format with **Transform OTU counts** (corrects for differences in sequencing depth)
- Results given as **log fold changes**

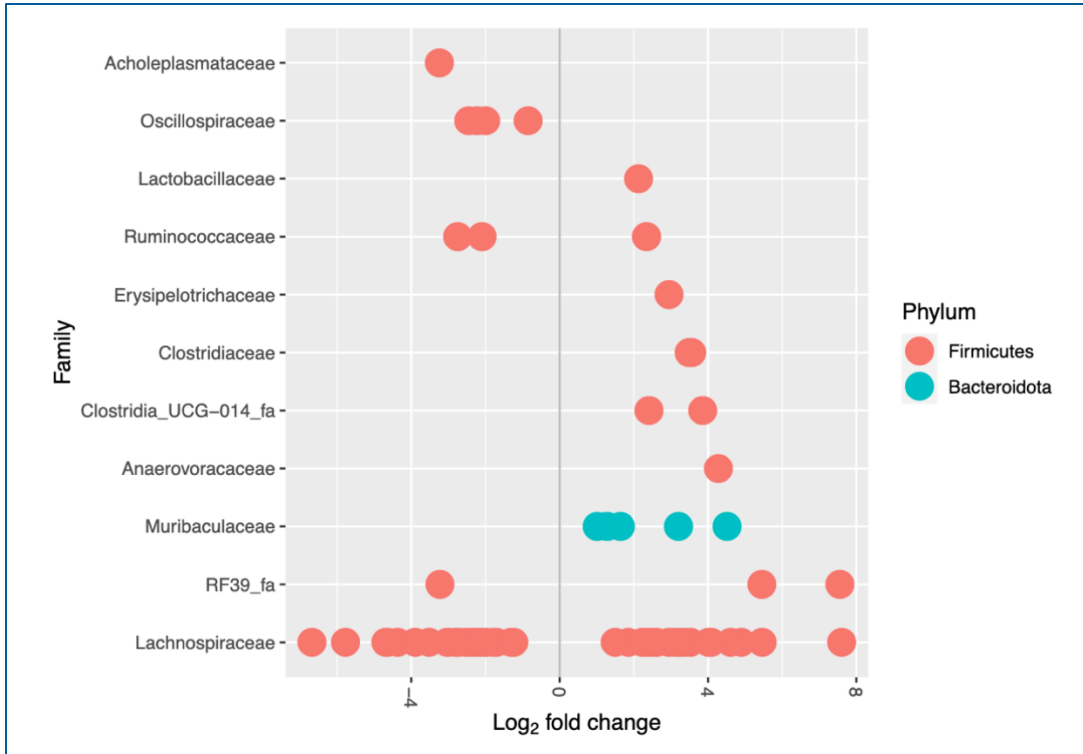
More information: <https://joey711.github.io/phyloseq-extensions/DESeq2.html>

DESeq2

Current tool configuration (April 2025):

- **Focused on comparison of two groups at a time**
 - If selected phenodata column has >2 groups, specify a pair (Group 1 and Group 2)
- **Reference level selection:**
 - Phenodata column with two groups -> first in alphabet is the reference level (e.g. 'b vs. a' or 'sick vs. healthy')
 - Phenodata column with >2 groups -> 'Group 2' is the reference level

DESeq2



8-fold increase compared to reference level = $\log_2$ fold change 3 (because $2^3=8$)

each dot = OTU with adjusted p value < 0.01

deseq2_otutable.tsv ...

Spreadsheet Text Volcano Plot Open in New Tab Details

Showing all 62 rows. [Full Screen](#)

	baseMean	log2FoldChange	lfcSE	padj	Phylum	Cla
Otu041	15.2682867002226	-6.67803203667824	0.702168578146934	1.84440147139632e-19	Firmicutes	Clostri
Otu049	16.056442049284	7.60372243170206	0.795369126187738	1.84440147139632e-19	Firmicutes	Clostri
Otu053	15.6531508754637	7.55398679303204	0.797443916253704	1.84440147139632e-19	Firmicutes	Bacilli
Otu026	24.2065517338278	-5.77349091671809	0.726321074599924	9.54591427870593e-14	Firmicutes	Clostri
Otu030	22.5693873723251	-3.03574845888934	0.41516376992914	1.06721315890685e-11	Firmicutes	Clostri
Otu018	47.8906882378487	3.52891895713938	0.509693715651345	1.27683063281188e-10	Firmicutes	Clostri
Otu022	30.1111553498981	-2.76523243473878	0.399307939457664	1.27683063281188e-10	Firmicutes	Clostri
Otu071	11.0106745359095	4.89494259540108	0.726736468385041	4.14508429055539e-10	Firmicutes	Clostri
Otu058	13.4115760907001	4.60767221203772	0.700695181336337	1.09108692505044e-09	Firmicutes	Clostri
Otu091	4.80679648598712	-4.68670014649616	0.745196451561208	6.47725981762776e-09	Firmicutes	Clostri
Otu095	4.70197614716378	-4.63581526833338	0.807724822781603	1.75393391653612e-07	Firmicutes	Clostri
Otu045	12.2846861028228	-2.74754726187511	0.490385518832907	3.56736342989387e-07	Firmicutes	Clostri
Otu119	3.68150112109875	5.45128296019317	0.9756277612781	3.59797615825661e-07	Firmicutes	Bacilli